A B Abhirama

+91 8660120678 | abhiram.b@icloud.com | portfolio.abhiram.tech | github.com/abhiramaab | Bengaluru, India

Summary

Self-taught backend developer building distributed services and event pipelines with **Java**, **Spring Boot**, and **Spring Security**. I fix bugs and contribute patches to large open-source codebases (**Apache Pulsar**, **Apache MINA SSHD**, **Spring Cloud**, **Apache BifroMQ**, **Spring Boot Admin**, **Apache Camel**, **Apache Shiro**) spanning authentication, Kafka transport, and concurrency. Before this I spent four years in supply chain and ERP operations.

Technical Skills

Languages & Frameworks: Java (8/17/21), Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate
Architecture & System Design: System Design (HLD & LLD), Idempotency, Distributed Locks, Transactional Outbox, Consistent Hashing, Double-Entry Ledger, Microservices, Spring Cloud
Databases & Caching: PostgreSQL, MySQL, Redis (Caching Strategies), Database Indexing, Query Optimization
DevOps & Tooling: Docker, Git, GitHub, Maven, AWS (EC2), Linux (Bash), Swagger/OpenAPI, Postman, JUnit 5, GitHub Copilot

Open Source Contributions

Featured pull requests in production distributed systems:

Apache Pulsar | *Distributed messaging and streaming platform* — PR #27022 (Merged)
- Fixed broker dispatch rate limiting overshoot under batched messages by dynamically scaling entry read limits using consumer batch estimates, preserving sub-batch forward progress, and handling uninitialized estimates before range filtering (fixes #26959, merged into 5.1.0).

Apache MINA SSHD | *Asynchronous SSH/SCP/SFTP protocol framework* — PR #953 (Merged)
- Resolved race condition in `IdentFilter` where premature CAS release of the write queue guard permitted concurrent messages (`SSH_MSG_KEX_FIRST`) to overtake queued `SSH_MSG_KEXINIT` packets; guarded queue release until outbound dispatch and added regression test for concurrent message ordering (fixes #952).

Spring Cloud Config | *Server-side config for distributed systems* — PR #3272
- Added passwordless Azure Managed Identity auth so Config Server can pull Git repos as a managed identity, on top of the existing PAT and SSH options. Acquires the token and injects it as a Bearer header into JGit's HTTP transport, falling back to the old path if unavailable.

Apache BifroMQ | *High-performance distributed MQTT broker* — PR #314 (Merged via #315)
- Diagnosed QoS 1 subscriber stall caused by Netty decoder vs. packet-sizer mismatch (Issue #313); authored fix aligning variable-length integer framing and drop-and-confirm delivery semantics, integrated into main via PR #315.

Apache Hop | *Data orchestration platform* — PR #9311 | #8344 (Merged)
- Fixed retry loop crash in RemoteHopServer by guarding zero/negative jitter bounds in backoff calculation (Issue #9278); resolved Hop Web stalled session bug via ServerPushSession keepalive (Issue #8327). Both merged.

Camel Kamelets | *Reusable connectors for Apache Camel* — PR #3039
- Kafka record keys were dropping whenever a message crossed an HTTP hop, which broke per-key ordering downstream. Mapped the standard kafka-key and CloudEvents ce-kafkakey headers through to CamelKafkaKey in the kafka-sink Kamelet.

Apache Shiro | *Authentication and authorization framework* — PR #2854
- Fixed a user principal name binding bug that broke Active Directory DN logins. Shipped in Shiro 3.0.1.

Spring Boot Admin | *Monitoring UI for Spring applications* — PR #5584
- Made the notification popup timeout configurable (it was hardcoded). Clicking a single-instance alert now jumps straight to that instance's detail page instead of the list. Backed by a Vitest case for the routing change.

Projects

LoomPay | *Java 21, Spring Boot, Redis, PostgreSQL, Docker, SOLID Design* — [Demo] | [Code] | [Docs]
- Built payment service with **Idempotency** (double-checked locking) and **Distributed Locks** (Redis SET NX EX) to stop duplicate charges during concurrent payment requests.
- Built a distributed **sliding-window rate limiter** using **Redis Sorted Sets (ZSET)** and a Spring **HandlerInterceptor** to throttle requests per merchant with **HTTP 429 Retry-After** headers, preventing downstream service overload during traffic spikes.
- Wired **Transactional Outbox Pattern** with a scheduled worker and retry limits so webhook events never drop when the network fails.
- Designed **Consistent Hashing ring router** with virtual nodes for load distribution, and **Double-Entry Ledger** for financial balance consistency.

Synoptiq | *Java, Spring Boot, Spring Security, PostgreSQL, Docker, AWS EC2* — [Demo] | [Docs]
- Workflow automation platform syncing Gmail and Google Calendar via OAuth2 and stateless JWT auth, storing records in PostgreSQL.
- Turned on HTTP/2 multiplexing so concurrent API calls and Server-Sent Events share one persistent TCP connection.
- Implemented background OAuth token refresh lifecycle and exponential backoff retry policies for Google API rate limits.

Routesphere | *Spring Boot, Spring Cloud, Eureka, Gateway, OpenFeign, Docker* — [Demo] | [Code] | [Docs]
- Logistics dispatch app split into 9 Spring Boot services covering orders, dispatch, and tracking.
- Eureka handles service discovery and Spring Cloud Gateway routes traffic with JWT validation done once at the edge.
- Services talk to each other over OpenFeign with client-side load balancing.

Experience

Supply Chain & ERP Operations — 2021 – 2026
Wipro Consumer Care, Medicon Leather, Mannol India, iThink Logistics — *Bengaluru, India*
- Ran SKU catalogs, order pipelines, and SAP ERP inventory reconciliation across 7 marketplaces including Amazon, Flipkart, and Myntra.
- Coordinated warehouse routing and regional fulfillment for a mid-size 3PL operation.

Education

University of Mysore — Expected 2028
Bachelor of Computer Applications (BCA), Online — *Mysuru, Karnataka*

Rural Institute of Advanced Studies — 2018
Higher Secondary Certificate (12th) — *Raipur, CG*